# OR1

Simple Harmonic Motion

November 13, 2025

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# 1 Disclaimer

The knowledge points of collation will inevitably have omissions and even fallacies • Written in a hurry, there is a high probability of omissions

# 2 Introduction

Based on previous enquiries, the midterm exam should cover modelling, simplex algorithms, dyadic theory and sensitivity analysis.

From my personal experience, modelling and simplex algorithms are the top priority.

Good luck!

# 3 Mathematical Formulation of LPP

*Decision variables*

- The quantities that need to be determined in order to solve the LPP are called decision variables.

*Objective function*

- The linear function of the decision variables, which is to be maximized or minimized, is called the objective function.

*Constraints*

- A constraint is something that plays the part of a physical, social, or financial restriction such as labor, machine, raw material, space, money, etc. These limits are the degrees to which an objective can be achieved.

*Sign restriction*

- If a decision variable $x_i$ can only assume nonnegative values, then we use the sign restriction $x_i \geq 0$. If a variable $x_i$ can assume positive, negative, or zero values, then we say that $x_i$ is unrestricted in sign.

## 3.1 General Linear Programming Problem

A general linear programming problem can be mathematically represented as follows:

$$\text{Maximize (or Minimize) } Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$$

subject to,

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1j}x_j + \ldots + a_{1n}x_n (\leq, =, \geq)\ b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2j}x_j + \ldots + a_{2n}x_n (\leq, =, \geq)\ b_2$$
$$\vdots$$
$$a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 + \ldots + a_{ij}x_j + \ldots + a_{in}x_n (\leq, =, \geq)\ b_i$$
$$\vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mj}x_j + \ldots + a_{mn}x_n (\leq, =, \geq)\ b_m$$

and $x_1, x_2, \ldots, x_n \geq 0$.

The above can be written in compact form as

$$\text{Maximize (or Minimize) } Z = \sum_{j=1}^{n} c_j x_j \quad (1.1)$$

subject to,

$$\sum_{j=1}^{n} a_{ij} x_j\ (\leq, =, \geq)\ b_i \quad \text{for } i = 1, 2, \ldots, m \quad (1.2)$$

$$x_j \geq 0 \quad \text{for } j = 1, 2, \ldots, n. \quad (1.3)$$

The problem is to find the values of $x_j$'s that optimize (maximize or minimize) the objective function (1.1). The values of $x_j$'s must satisfy the constraints (1.2) and non-negativity restrictions (1.3). Here, the coefficients $c_j$'s are referred to as cost coefficients and $a_{ij}$'s as technological coefficients; $a_{ij}$ represents the amount of the $i$th resource consumed per unit variable $x_j$, and $b_i$, the total availability of the $i$th resource.

**Example** An oil company owns two refineries – refinery A and refinery B. Refinery A is capable of producing 20 barrels of petrol and 25 barrels of diesel per day. Refinery B is capable of producing 40 barrels of petrol and 20 barrels of diesel per day. The company requires at least 1000 barrels of petrol and at least 800 barrels of diesel. If it costs Rs. 300 per day to operate refinery A and Rs. 500 per day to operate refinery B, how many days should each refinery be operated by the company so as to minimize costs? Formulate this problem as a linear programming model.

**Solution:** Let $x_1$ and $x_2$ be the numbers of days the refineries A and B are to be operated, respectively. Our objective is to minimize $Z = 300x_1 + 500x_2$. The total amount of petrol produced is $20x_1 + 40x_2$. As at least 1000 barrels of petrol are

required, we have the inequality $20x_1 + 40x_2 \geq 1000$. Similarly, the total amount of diesel produced is $25x_1 + 20x_2$. As at least 800 barrels of diesel are required, we have the inequality $25x_1 + 20x_2 \geq 800$. Hence our linear programming model is

$$\begin{aligned}
&\text{Minimize } Z = 300x_1 + 500x_2 \\
&\text{subject to} \\
&\qquad 20x_1 + 40x_2 \geq 1000 \\
&\qquad 25x_1 + 20x_2 \geq 800 \\
&\qquad x_1, x_2 \geq 0.
\end{aligned}$$

## 3.2 LP Solution

### 3.2.1 Some Terminologies for Solution

Firstly some basic concepts.

• Convex set

A set is convex if and only if, for any two points on the set, the line segment joining those two points lies entirely in the set. Mathematically, a set $S$ is said to be convex if for all $x, y \in S$, $\lambda x + (1 - \lambda)y \in S$, for all $\lambda \in [0, 1]$.

• Extreme points

The extreme points of a convex polygon are the points of intersection of the lines bounding the feasible region.

• Feasible Solution(FS)

Any non-negative solution which satisfies all the constraints is known as a feasible solution of the problem.

• Basic solution (BS)

- For a set of $m$ simultaneous equations in $n$ variables ($n > m$) in an LP problem, a solution obtained by setting $(n-m)$ variables equal to zero and solving for the remaining $m$ equations with $m$ variables is called a basic solution. These $m$ variables are called basic variables and $(n - m)$ variables are called non-basic variables.

• Basic feasible solution (BFS)

- A basic solution to an LP problem is called a basic feasible solution (BFS) if it satisfies all the non-negativity restrictions. A BFS is called degenerate if the value of at least one basic variable is zero, and non-degenerate if the values of all basic variables

are non-zero and positive.

- Optimal basic feasible solution

- A basic feasible solution is called optimal if it optimizes (maximizes or minimizes) the objective function.

#### 3.2.2 Graphical Method

### 3.3 Simplex Method

#### 3.3.1 Notes

- Artificial variable
- M

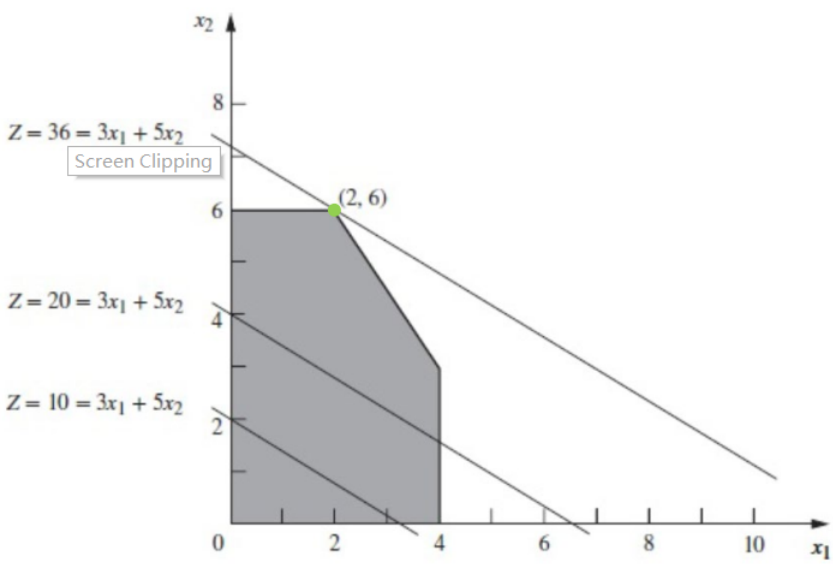


Figure 1: Illustration of LP

#### 3.3.2 Simplex Algorithm

For the solution of any LP problem by simplex algorithm, the existence of an initial BFS is always assumed.

**From the textbook of Prof. Bibhas C. Giri**

**Step 1:** Convert the objective function into maximization type if the given LPP is of minimization type. Also, convert all the $m$ constraints such that $b_i$'s $(i = 1, 2, ..., m)$ are all non-negative. Then convert each inequality constraint into an equation by introducing slack or surplus variable and assign a zero cost coefficient to such a variable in the objective function.

**Step 2:** If needed, introduce artificial variable(s) and take $(-M)$ as the coefficient of each artificial variable in the objective function.

**Step 3:** Obtain the initial basic feasible solution $xB = B^{-1}b$ where $B$ is the basis matrix, which is an identity matrix here.

**Step 4:** Calculate the net evaluations $zj - cj = c_B x_B - c_j$.

(i) If $zj - cj \geq 0$ for all $j$, then $xB$ is an optimum BFS.

(ii) If at least one $zj - cj < 0$, then proceed to improve the solution in the next step.

**Step 5:** If there are more than one negative $zj - cj$, then choose the most negative of them. Let it be $zk - ck$ for some $j = k$.

(i) If all $a_{ik} < 0$ $(i = 1, 2, ..., m)$, then there exists an unbounded solution to the given problem.

(ii) If at least one $a_{ik} > 0$ $(i = 1, 2, ..., m)$, then the corresponding vector $a_k$ enters the basis $B$. This column is called the key or pivot column.

**Step 6:** Divide each value of $xB$ (i.e., $b_i$) by the corresponding (but positive) number in the key column and select a row which has the ratio non-negative and minimum, i.e.,

$$x_{Br}/a_{ik} > 0$$

This rule is called the minimum ratio rule. The row selected in this manner is called the key or pivot row and it represents the variable which will leave the basic solution. The element that lies in the intersection of the key row and key column of the simplex table is called the key or pivot element (say $a_{rk}$).

**Step 7:** Convert the leading element to unity by dividing its row by the key element itself and all other elements in its column to zeros by making use of the relation:

$$\hat{a}_{rj} = \frac{a_{rj}}{a_{rk}} \quad \text{and} \quad \hat{x}_B^{(r)} = \frac{x_{Br}}{a_{rk}}, \quad i = r; j = 1, 2, ..., n$$

$$\hat{a}_{ij} = a_{ij} - \frac{a_{rj}}{a_{rk}} a_{ik} \quad \text{and} \quad \hat{x}_{Bi} = x_{Bi} - \frac{x_{Br}}{a_{rk}} a_{ik}, \quad i = 1, 2, ...., m; i \neq r$$

**Step 8:** Go to step 4 and repeat the procedure until all entries in $(zj - cj)$ are either positive or zero, or there is an indication of an unbounded solution.

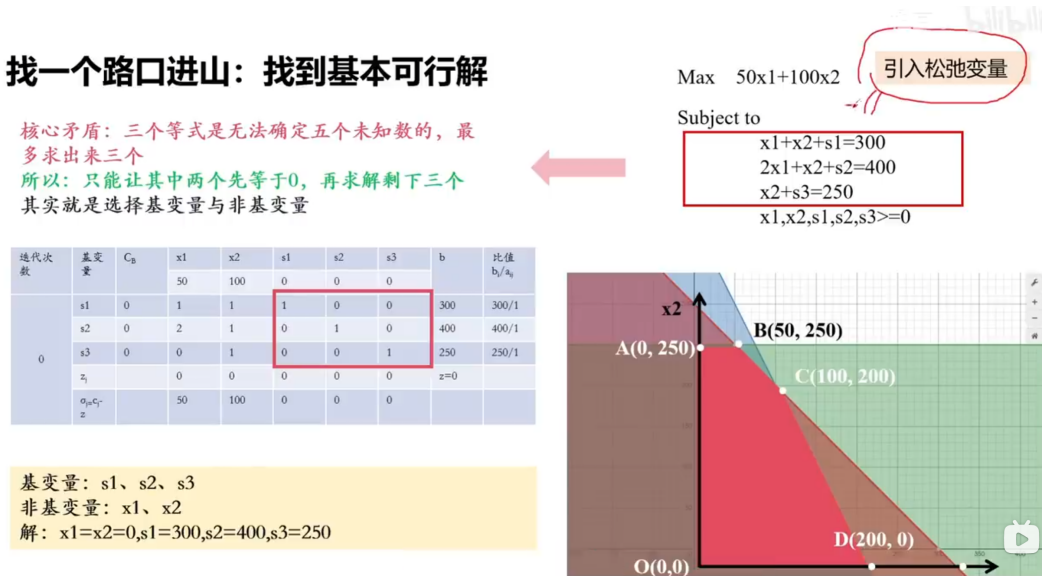


Figure 2: Illustration of Simplex

#### 3.3.3 Simplex Table

**Example 2.2:**

$$\begin{aligned} \text{maximize} \quad & 2x_1 - x_2 + 2x_3 \\ \text{subject to} \quad & 2x_1 + x_2 = 10 \\ & x_1 + 2x_2 - 2x_3 = 25 \\ & x_1 + 2x_2 = 5 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0 \end{aligned}$$

**Solution:** Using the basic feasible solution $(x_1, x_2, x_3, s_1, s_2, s_3) = (0, 0, 0, 10, 25, 5)$, the initial simplex tableau for this problem is as follows:

| $z$ | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $b$ |
|---|---|---|---|---|---|---|---|
| 1 | $-2$ | 1 | $-2$ | 0 | 0 | 0 | 0 |
| 0 | 2 | 1 | 0 | 1 | 0 | 0 | 10 |
| 0 | 1 | 2 | $-2$ | 0 | 1 | 0 | 25 |
| 0 | 0 | 1 | 2 | 0 | 0 | 1 | 5 |

And the optimal tableau is:

| $z$ | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $b$ |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 3 | 0 | 1 | 0 | 1 | 15 |
| 0 | 1 | 0.5 | 0 | 0.5 | 0 | 0 | 5 |
| 0 | 0 | 2.5 | 0 | $-0.5$ | 1 | 1 | 20 |
| 0 | 0 | 0.5 | 1 | 0.5 | 0 | 1 | 2.5 |

This implies that the optimal solution is $(x_1, x_2, x_3, s_1, s_2, s_3) = (5, 0, \frac{5}{2}, 0, 25, 0)$, and the maximum value of $z$ is 15.

## 4 Artificial Variable and Solve Methods

In the standard form of an Linear Programming Problem (LPP), when the decision variables together with slack and surplus variables cannot afford initial basic variables, a non-negative variable is introduced to the left-hand side (LHS) of each equation which lacks the starting basic variable. This variable is known as an artificial variable.

The Big M method is a technique for solving LPPs that involve artificial variables. In this method, a very large negative price $(-M)$ (where $M$ is positive) is assigned to each artificial variable in the objective function of maximization type. After introducing the artificial variable(s), the problem can be solved using the usual simplex method. However, while solving in simplex, the following conclusions are drawn from the final table:

(a) The current solution is an optimal Basic Feasible Solution (BFS) if no artificial variable remains in the basis, and the optimality condition is satisfied.

(b) The current solution is an optimal degenerate BFS if at least one artificial variable appears in the basis at zero level, and the optimality condition is satisfied.

(c) The problem has no feasible solution if at least one artificial variable appears in the basis at a positive level, and the optimality condition is satisfied.

**Problem 17**

用大 $M$ 法求解下述线性规划问题:

$$\begin{aligned} \min\ & 2x_1 - x_2 + 2x_3 \\ \text{s.t.}\ & x_1 + 3x_2 - 2x_3 \le 8 \\ & 2x_1 + x_2 \ge 3 \\ & 3x_1 + x_2 - x_3 \ge 5 \\ & x_1,\ x_2,\ x_3 \ge 0 \end{aligned}$$

Figure 3: Big M Method

**Problem 6**

请将下面的线性规划问题转化为标准形式：

$$
\begin{aligned}
\min\ & z = x_1 - 2x_2 + 3x_3 \\
\text{s.t.}\ & x_1 + x_2 + x_3 \le 6 \\
& x_1 - x_2 + x_3 \ge 3 \\
& -3x_1 + x_2 + 2x_3 = -4 \\
& x_1 \ge 0,\ x_2 \ge 0
\end{aligned}
$$

Figure 4: Two Stage Method

There is also a two-stage approach. The first stage starts by minimising the artificial variables and if there is a solution in the first stage an initial feasible solution is obtained in the second stage.

# 5 Some Special Cases in LPP

## 5.1 Unrestricted Variables

Sometimes, decision variables are unrestricted in sign, meaning they can take positive, negative, or zero values. In such cases, the decision variables can be expressed as the difference between two non-negative variables. For example, if $x_1$ is unrestricted in sign, then it can be expressed as the difference between two non-negative variables, $x_1'$ and $x_1''$, such that $x_1 = x_1' - x_1''$.

## 5.2 Unbounded Solution

In general, an unbounded solution can be detected at any iteration if all the coefficients $a_{ij} \leq 0$ for $i = 1, 2, \ldots, m$ in the entering vector $\mathbf{a}_j$ for which $z_j - c_j < 0$.

## 5.3 No Feasible Solution

If at any stage of simplex computation, one or more artificial variables remain in the basis at positive level then the problem has no feasible solution.

# 6 Duality

Every Linear Programming Problem (LPP) is associated with another unique LPP. The original problem is called the "primal," while the other one is called the "dual." In fact, either problem can be considered as primal and the remaining problem as its dual.

An interesting feature of duality is that if the optimal solution of one is known, then the optimal solution of the other one is readily available. If the primal problem contains a large number of constraints and a few decision variables, then solving the dual problem is the ideal case to obtain the solution of the primal, as it can reduce computational burden.

**Weak Duality:** The optimal value of the objective function of the primal problem is always greater than or equal to the optimal value of the objective function of the dual problem. This property ensures that if one problem has a finite optimal solution, the other cannot have an unbounded solution.

**Strong Duality:** If both the primal and the dual have feasible solutions, then they have the same optimal objective function value (assuming they are bounded). This

means that the optimal value of the primal problem equals the optimal value of the dual problem.

**Complementary Slackness:** A condition in linear programming that provides a connection between the optimal solutions of the primal and dual problems. It states that for each pair of primal and dual constraints, at least one must be "slack" (i.e., not binding or not fully utilized) when the optimal solutions are reached. Formally:

1. If a primal constraint $A_i x \leq b_i$ is not binding (i.e., $A_i x < b_i$), then the corresponding dual variable $\pi_i$ must be zero.

2. Conversely, if the dual constraint $A_i^T \pi \geq c_i$ is not binding, then the corresponding primal variable $x_i$ must be zero.

### 6.1 Mathematical Formulation of Dual

Suppose that a Linear Programming Problem (LPP) is given in standard form as

$$\text{Maximize } Z = cx \quad \text{subject to } Ax \leq b, \text{ and } x \geq 0$$

where $A$ is an $m \times n$ matrix, $b^T \in R^m$, and $c, x^T \in R^n$.

Then the associated LPP

$$\text{Minimize } W = b^T v \quad \text{subject to } A^T v \geq c^T, \text{ and } v \geq 0$$

where $v \in R^m$, is called the dual of the original (primal) problem.

Notice that $\min Z_1 = \max(-Z_1) = -\max Z_1 = cx = \max Z$ and $-Ax \geq -b$ is equivalent to $Ax \leq b$.

Hence the dual problem can be written as

$$\text{Maximize } Z = cx \quad \text{subject to } Ax \leq b, \text{ and } x \geq 0$$

**Corollary 1** *If either the primal or dual problem has an unbounded objective function value, then the other problem has no feasible solution.*

### 6.2 Duality in Simplex Method

Since any Linear Programming (LP) problem can be solved by the simplex method, the method is applicable to both the primal as well as its dual.

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \geq 0. \end{aligned}$$

| Max | | Min | |
|---|---|---|---|
| Constraint | <= bi | >= 0 | Variable |
| | = bi | Free | |
| | >= bi | <= 0 | |
| Variable | >= 0 | >= cj | Constraint |
| | Free | = cj | |
| | <= 0 | <= cj | |

Figure 5: Primal-Dual Relationship

The fundamental theorem of duality enables us to find the optimal solution of the dual (primal) by reading the solution of the primal (dual), as given below:

1. If the primal (dual) variable is related to slack and/or surplus variables in the dual (primal) problem, then its optimal solution can be read off from the net evaluation row of the optimal dual (primal) simplex table as the net evaluation corresponding to the slack and/or surplus variable.

2. If the primal (dual) variable is related to an artificial starting variable in the dual (primal) problem, then its optimal solution is read off from the net evaluation row of the optimal dual (primal) simplex table as the net evaluation related to this artificial variable after putting the penalty cost $M = 0$ and changing the sign of net evaluations.

3. If either the primal or dual has an unbounded solution, then the other one has no feasible solution.

#### 6.2.1 Dual Simplex Case

Meijia Company plans to manufacture two types of household electrical products, denoted as Product I and Product II. The company has information about the time required by Equipment A, Equipment B, and the testing workshop per unit of each product, as well as the daily capacity available for these resources. The profit earned from selling one unit of each product is also provided. The data is shown in Table 1-1.

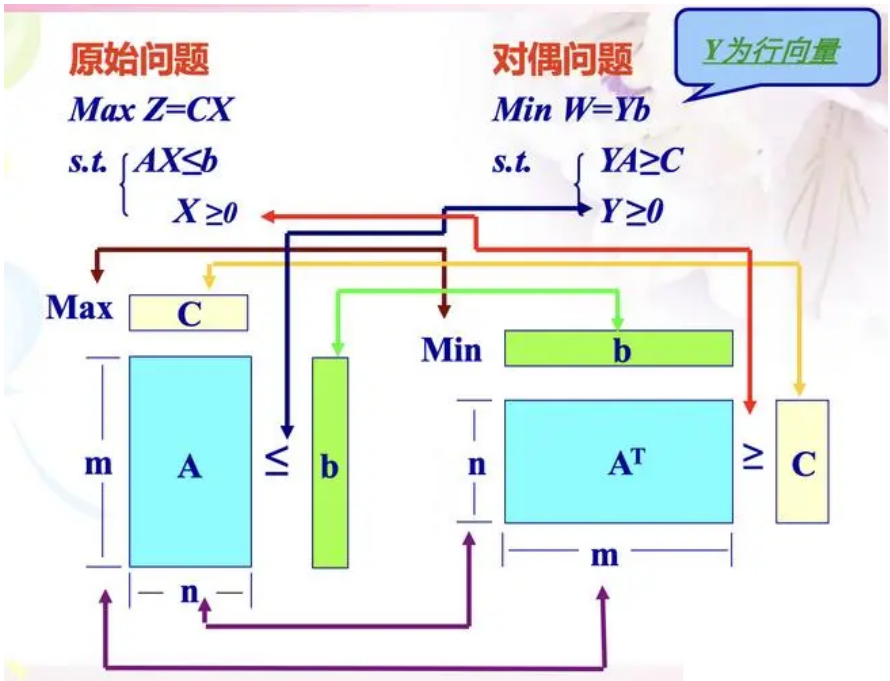


Figure 6: Dual Problem

The question is: How many units of each product should the company manufacture daily to maximize profit?

| Item | Product I | Product II | Daily Available Capacity |
|---|---|---|---|
| Equipment A (hours/unit) | 0 | 5 | 15 |
| Equipment B (hours/unit) | 6 | 2 | 24 |
| Testing Workshop (hours/unit) | 1 | 1 | 5 |
| Profit (¥/unit) | 2 | 1 | - |

Table 1: Production data for Products I and II

Solution.

### 6.3 KKT Conditions (Intuitive Version)

We want to solve

$$\min f(x)$$

subject to

$$g_i(x) \leq 0 \quad (i = 1, \ldots, m), \qquad h_j(x) = 0 \quad (j = 1, \ldots, p).$$

Introduce multipliers:

$$\lambda_i \geq 0 \text{ for } g_i(x) \leq 0, \qquad \nu_j \in R \text{ for } h_j(x) = 0.$$

Define the Lagrangian

$$L(x, \lambda, \nu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \nu_j h_j(x).$$

A point $x^*$ is (locally) optimal only if there exist $(\lambda^*, \nu^*)$ such that the following KKT conditions hold:

1. **Stationarity**

$$\nabla_x L(x^*, \lambda^*, \nu^*) = \nabla f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^{p} \nu_j^* \nabla h_j(x^*) = 0.$$

2. **Primal feasibility**

$$g_i(x^*) \leq 0, \quad h_j(x^*) = 0.$$

3. **Dual feasibility**

$$\lambda_i^* \geq 0.$$

4. **Complementary slackness**

$$\lambda_i^* g_i(x^*) = 0 \quad \text{for each } i.$$

*Meaning of complementary slackness:* for each constraint $g_i(x) \leq 0$,

$$\lambda_i^* > 0 \;\Rightarrow\; g_i(x^*) = 0 \text{ (active)}, \qquad g_i(x^*) < 0 \;\Rightarrow\; \lambda_i^* = 0 \text{ (no penalty)}.$$

These four short rules are the KKT conditions: they generalize the Lagrange multiplier idea to handle both $\leq$-constraints and $=$-constraints.

# 7 Sensitivity Analysis

In real-world applications, the parameters and coefficients in LP models are often subject to uncertainty and variability.

Sensitivity analysis plays a crucial role in understanding the robustness and reliability of linear programming solutions when faced with changes in the input data. It involves examining how changes in the coefficients of the objective function, right-hand side values of constraints, or both, impact the optimal solution and the optimal value of the objective function.

Note: There doesn't seem to be a critical test in our session. Or at least it wasn't too hard.

灵敏度分析的步骤可归纳如下：

1. 将参数的改变通过计算反映到最终单纯形表上来。

具体计算方法是，按下列公式计算出由参数 $a_{ij}$，$b_i$，$c_j$ 的变化而引起的最终单纯形表上有关数字的变化。由式(2.12)、式(2.13)和式(2.17)可得出下列各式：

$$\Delta \boldsymbol{b}' = \boldsymbol{B}^{-1}\Delta \boldsymbol{b} \tag{2.30}$$

$$\Delta \boldsymbol{P}_j' = \boldsymbol{B}^{-1}\Delta \boldsymbol{P}_j \tag{2.31}$$

$$(c_j - z_j)' = c_j - \sum_{i=1}^{m} a_{ij} y_i' \tag{2.32}$$

2. 检查原问题是否仍为可行解。
3. 检查对偶问题是否仍为可行解。
4. 按表 2-9 所列情况得出结论或决定继续计算的步骤。

表 2-9

| 原问题 | 对偶问题 | 结论或继续计算的步骤 |
|---|---|---|
| 可行解 | 可行解 | 问题的最优解或最优基不变 |
| 可行解 | 非可行解 | 用单纯形法继续迭代求最优解 |
| 非可行解 | 可行解 | 用对偶单纯形法继续迭代求最优解 |
| 非可行解 | 非可行解 | 引进人工变量，编制新的单纯形表重新计算 |

Figure 7: Sensitivity Analysis

# 8 Some other notes

## 8.1 Formatting

Be careful not to have non-linear parts such as multiplication when modelling.

## 8.2 Standard and general forms

The main difference is between an equal sign and an unequal sign.

## 8.3 Conditions

This section must be memorised carefully.

#### 8.3.1 Special Constraints

**Either-Or Constraints:**

We want to ensure that either $f(x_1, x_2, \ldots, x_n) \leq 0$ or $g(x_1, x_2, \ldots, x_n) \leq 0$ is satisfied. This can be represented as:

$$g(x_1, x_2, \ldots, x_n) \leq M \cdot y,$$
$$f(x_1, x_2, \ldots, x_n) \leq M \cdot (1 - y),$$

where $y \in \{0, 1\}$ is a binary variable and $M$ is a sufficiently large constant. This formulation ensures that: - When $y = 0$, $f(x_1, x_2, \ldots, x_n) \leq 0$ must hold. - When $y = 1$, $g(x_1, x_2, \ldots, x_n) \leq 0$ must hold.

**If-Then Constraints:**

For an if-then constraint, where you want $f(x_1, x_2, \ldots, x_n) > 0$ to imply $g(x_1, x_2, \ldots, x_n) \geq 0$, we can also use a binary variable $y$ and a large constant $M$ to model it.

If we want $f(x_1, x_2, \ldots, x_n) > 0$ to imply $g(x_1, x_2, \ldots, x_n) \geq 0$, we can use the following formulation:

$$-g(x_1, x_2, \ldots, x_n) \leq M \cdot y,$$
$$f(x_1, x_2, \ldots, x_n) \leq M \cdot (1 - y),$$

where $y \in \{0, 1\}$ is a binary variable and $M$ is a sufficiently large constant. This ensures that: - When $f(x_1, x_2, \ldots, x_n) > 0$, we set $y = 0$, forcing $g(x_1, x_2, \ldots, x_n) \geq 0$. - When $f(x_1, x_2, \ldots, x_n) \leq 0$, $g(x_1, x_2, \ldots, x_n) \geq 0$ is optional, allowing $y = 1$.

These formulations require selecting a sufficiently large constant $M$ such that $M$ does not constrain the feasible region unless required by the logic.

### 8.4 Some Problems

#### 8.4.1 Mixed Material

A certain fruit candy factory uses raw materials A, B, and C to produce three different brands of candy, labeled as Brand Jia, Brand Yi, and Brand Bing. The available data includes the proportions of each raw material in each brand, the cost per kilogram of each raw material, monthly supply limits for each raw material, and the processing and selling price per kilogram for each brand. This information is provided in Table 1-17.

The question is: How many kilograms of each brand should be produced each month to maximize profit? Formulate the linear programming model for this problem.

| **Raw Material** | **Brand Jia** | **Brand Yi** | **Brand Bing** | **Cost of Raw Material** | **Monthly Supply Limit** |
|---|---|---|---|---|---|
| A | $\geq 60\%$ | - | $\leq 30\%$ | 2.00 | 2,000 |
| B | - | $\leq 50\%$ | $\leq 60\%$ | 1.50 | 2,500 |
| C | $\leq 20\%$ | $\geq 50\%$ | - | 2.70 | 1,200 |
| Processing Cost | 0.50 | 0.40 | 0.30 | - | - |
| Selling Price | 3.40 | 2.85 | 2.25 | - | - |

Table 2: Production data for Brands Jia, Yi, and Bing

#### 8.4.2 Scheduling

The Haicheng Public Transport Company is responsible for the operation and maintenance management of all public buses in the city. The required number of personnel for each time period is shown in Table 3. Every employee works a 6-hour shift starting at different times throughout the day, with each employee receiving at least one day off each week.

Questions:

1. How many employees does the company need to deploy to ensure normal operations?

2. Given the staggered work hours, with some employees starting as early as 5:00 AM and others finishing as late as 11:00 PM, along with weekly rest days (some on weekends, some during the week), propose an arrangement for shift schedules, ensuring each employee has one rest day within the workweek. Consider balancing the needs of the company and the well-being of the employees. How can a fair schedule be achieved?

| **Time** | 5:00–7:00 | 7–9 | 9–11 | 11–13 | 13–15 | 15–17 | 17–19 | 19–21 | 21–23 |
|---|---|---|---|---|---|---|---|---|---|
| **Required** | 26 | 50 | 34 | 40 | 38 | 50 | 46 | 36 | 26 |

Table 3: Personnel requirements for each time period

### 8.4.3 Transportation

本章研究单一品种物资的运输调度问题，其典型情况是：设某种物品有 $m$ 个产地 $A_1, A_2, \cdots, A_m$，各产地的产量分别是 $a_1, a_2, \cdots, a_m$；有 $n$ 个销地 $B_1, B_2, \cdots, B_n$，各销地的销量分别为 $b_1, b_2, \cdots, b_n$。假定从产地 $A_i (i=1,2,\cdots,m)$ 向销地 $B_j (j=1,2,\cdots,n)$ 运输单位物品的运价是 $c_{ij}$，问怎样调运这些物品才能使总运费最小？

为直观清楚起见，可列出该问题的运输表，如表 3-1 所示。表中的变量 $x_{ij} (i=1,2,\cdots,m;\ j=1,2,\cdots,n)$ 为由产地 $A_i$ 运往销地 $B_j$ 的物品数量，$c_{ij}$ 为 $A_i$ 到 $B_j$ 的单位运价。有时，将单位运价单独列入另一个表中，并称其为单位运价表。

**表 3-1**

| 产地 \ 销地 | $B_1$ | $B_2$ | $\cdots$ | $B_n$ | 产量 |
|---|---|---|---|---|---|
| $A_1$ | $c_{11}$ $x_{11}$ | $c_{12}$ $x_{12}$ | | $c_{1n}$ $x_{1n}$ | $a_1$ |
| $A_2$ | $c_{21}$ $x_{21}$ | $c_{22}$ $x_{22}$ | | $c_{2n}$ $x_{2n}$ | $a_2$ |
| $\vdots$ | $\vdots$ | $\vdots$ | | $\vdots$ | $\vdots$ |
| $A_m$ | $c_{m1}$ $x_{m1}$ | $c_{m2}$ $x_{m2}$ | | $c_{mn}$ $x_{mn}$ | $a_m$ |
| 销量 | $b_1$ | $b_2$ | $\cdots$ | $b_n$ | |

Figure 8: Transportation Problem

**例 1** 某部门有 3 个生产同类产品的工厂（产地），生产的产品由 4 个销售点（销地）出售，各工厂的生产量、各销售点的销售量（假定单位均为 t）以及各工厂到各销售点的单位运价（元/t）示于表 3-2 中。要求研究产品如何调运才能使总运费最小。

**表 3-2**

| 产地 \ 销地 | $B_1$ | $B_2$ | $B_3$ | $B_4$ | 产量 |
|---|---|---|---|---|---|
| $A_1$ | 4 | 12 | 4 | 11 | 16 |
| $A_2$ | 2 | 10 | 3 | 9 | 10 |
| $A_3$ | 8 | 5 | 11 | 6 | 22 |
| 销量 | 8 | 14 | 12 | 14 | 48 |

Figure 9: Case Problem

### 8.4.4 Complex Modeling

A company has received an order to deliver 20,000 radio units within four weeks. The client is willing to pay \$20 per unit if delivered before the end of the first week, \$18 per unit if delivered before the end of the second week, \$16 per unit if delivered before the end of the third week, and \$14 per unit if delivered before the end of the fourth week. Each worker can assemble 50 units per week, but the current workforce of 40 workers is insufficient to meet the order volume.

Therefore, the company needs to hire and train temporary helpers. Any experienced worker can supervise up to 3 trainees, and each trainee can enter the assembly line or be upgraded to a higher level weekly. The company does not have contracts with

other companies, so once these helpers are hired, they must remain on the payroll until the end of the fourth week, regardless of their status (permanent or temporary). The weekly salary for each worker, whether regular or trainee, is \$200; for each trainee, it is \$100. The production cost per unit (excluding labor costs) is \$5.

The company is considering the following weekly production plan to maximize profit, shown in detail below:

- **Week 1:** 10 assembly workers, 30 instructors, 90 trainees. Worker wage: \$8,000. Trainee wage: \$9,000. Profit from 500 units: \$7,500. Loss: \$9,500.
- **Week 2:** 120 assembly workers, 10 instructors, 30 trainees. Worker wage: \$26,000. Trainee wage: \$3,000. Profit from 6,000 units: \$78,000. Net profit: \$49,000.
- **Week 3:** 160 assembly workers. Worker wage: \$32,000. Profit from 8,000 units: \$88,000. Net profit: \$56,000.
- **Week 4:** 110 assembly workers, 50 idle. Worker wage: \$32,000. Profit from 5,500 units: \$49,500. Net profit: \$17,500.

This plan is one of many possible scenarios, with a total net profit of \$113,000. The company's goal is to maximize the total net profit. Formulate this problem as an LP (Linear Programming) model.

#### 8.4.5 0-1 Variable

A company plans to relocate a portion of its production vehicles from its main city area to two satellite towns, labeled as Town Jia and Town Yi. These towns offer advantages in terms of land costs, rent, and waste disposal costs, which are relatively inexpensive. However, relocating to these towns will increase transportation costs between locations. The company has five vehicles, labeled A, B, C, D, and E, operating in the main city area. The relocation plan limits the number of vehicles stationed in each satellite town (Town Jia and Town Yi) to a maximum of 3.

The annual transportation cost of moving vehicles from the main city to the satellite towns is provided in Table C-18.

However, additional transportation costs are incurred due to the relocation, which depend on $C_{ik}$ and $D_j$ values. Here, $C_{ik}$ represents the annual transportation volume (tons/year) between vehicles $i$ and $k$, and $D_j$ represents the unit transportation cost

| Vehicle | A | B | C | D | E |
|---|---|---|---|---|---|
| Relocation to Town Jia | 100 | 150 | 100 | 200 | 50 |
| Relocation to Town Yi | 100 | 200 | 150 | 150 | 150 |

Table 4: Annual relocation costs from the main city to satellite towns (¥10,000/year)

per ton from the main city area to the satellite towns. The specific data for $C_{ik}$ and $D_j$ are provided in Tables C-19 and C-20.

| Vehicle | B | C | D | E |
|---|---|---|---|---|
| A | 0 | 1000 | 1500 | 0 |
| B | - | 1400 | 1200 | 0 |
| C | - | - | 0 | 2000 |
| D | - | - | - | 700 |

Table 5: $C_{ik}$ Values (tons/year) between vehicles

| Location | Town Jia | Town Yi | Main City Area |
|---|---|---|---|
| Jia | 500 | 1400 | 1300 |
| Yi | - | 500 | 900 |
| Main City Area | - | - | 1000 |

Table 6: $D_j$ Values (¥/ton) for unit transportation cost to various locations

The task is to determine the optimal vehicle relocation plan to minimize transportation costs.

#### 8.4.6 Discussion of LP

1. 考虑线性规划问题

$$\begin{aligned} \max \quad & z = 4x_1 - x_2 + 2x_3, \\ \text{s.t.} \quad & x_1 + x_2 \leq 5, \\ & 2x_1 + x_2 \leq 7, \\ & 2x_2 + x_3 \geq 6, \\ & x_1 + x_3 = 4, \\ & x_1 \geq 0, x_2 \geq 0, x_3 \in \mathbb{R}, \end{aligned} \tag{1.1}$$

试回答如下问题：

(a) 判断 $x^* = (\frac{8}{3}, \frac{7}{3}, \frac{4}{3})^T$ 是否是该问题的最优解？

(b) 试利用单纯形算法，求出该问题的最优解和最优值。

(c) 该问题的最优解是否惟一？为什么？

Figure 10: Case from AOR1

# 9 Appendix

Hungarian algorithm

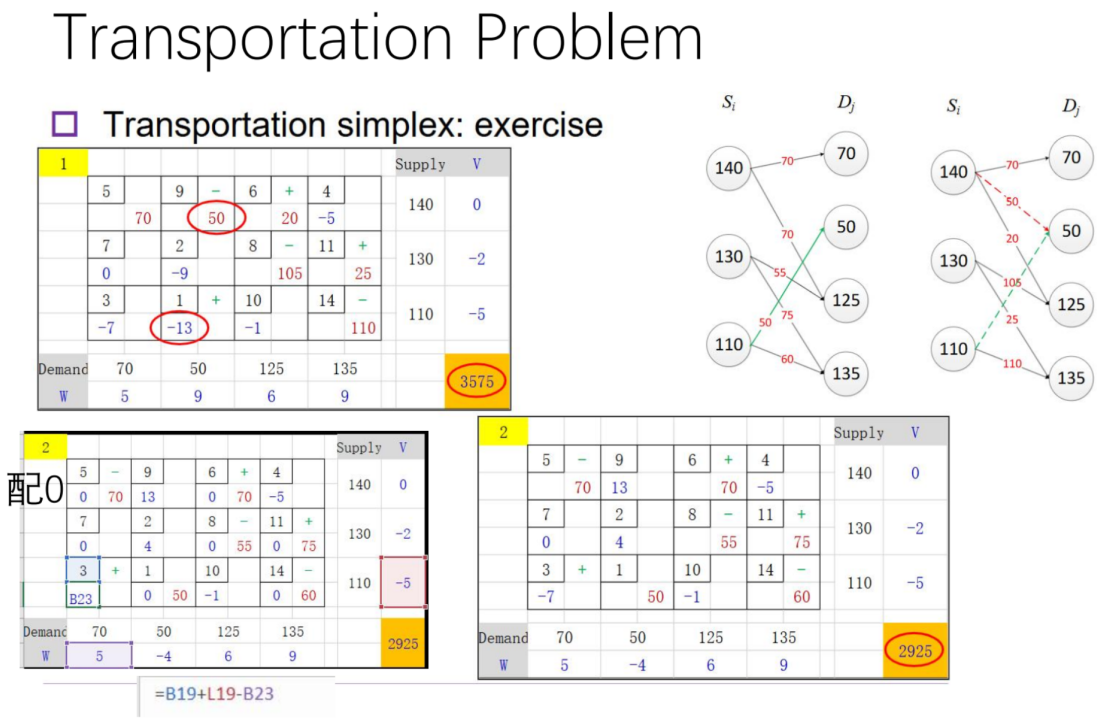


Figure 11: Hungarian algorithm

## 10 Special Thanks

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